

H.W. 27-07-26

Trace quicksort as shown in the class
the following array.

[85 76 99 13 45 103 -5 26 31]

350	790	400	200	100	96	9000	890
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[760]

1) 85 76 99 13 45 103 -5 26 31

$a[i]$

$a[j]$

pivot

$a[i] < \text{pivot}$

$a[j] > \text{pivot}$

85 76 99 13 45 103 -5 26 31

~~i~~ ~~i~~ i

j

swap i & j

85 76 31 13 45 103 -5 26 99

~~i~~ ~~i~~ i ~~i~~ ~~i~~ ~~i~~ j ~~i~~

swap i & j

85 76 31 13 45 26 -5 103 99

i / j / / / / / i /

swap pivot & j

j /

-5 | 76 31 13 45 26 85 103 99

/ / / / / i /

j /

swap pivot & j

-5 | 26 31 76 45 76 85 103 99

/ i / / / / /

swap i & j

-5 | 26 13 31 45 76 85 103 99

/ / i / / / /

i /

swap pivot & j

-5 | 13 26 31 45 76 85 103 99
-5 13 26 31 45 76 85 | 103 99

swap i & j

-5 13 26 31 45 76 85 99 103

i	j	a[i]	a[j]	a[pivot]
0	8	85	31	85
5	7	103	31	85

i	j	a[i]	a[j]	a[pivot]
7	6	103	-5	85
6	5	85	26	85
1	6	76	85	76
6	5	85	26	76
1	5	85	26	85
2	2	31	31	31
6	5	103	85	85
1	2	26	81	26
7	8	99	103	103

② 350 790 400 200 100 96 900 890 760
~~i~~ ~~j~~ ~~j~~ ~~j~~ ~~j~~

Swap i & j

350 96 400 200 100 790 900 890 760
~~i~~ ~~j~~ ~~j~~ ~~j~~ ~~j~~ ~~j~~ ~~j~~ ~~j~~

Swap i & j

350 96 100 200 400 790 900 890 760
~~i~~ ~~j~~ ~~j~~ ~~j~~ ~~j~~ ~~j~~ ~~j~~ ~~j~~

Swap pivot & j

200 96 100 | 350 | 400 790 900 890 760

~~i~~ ~~j~~ i j

100 96 200 | 350 | 400 790 900 890 760

~~i~~ ~~j~~ i j ~~i~~ i ~~j~~ ~~j~~ ~~j~~
i j i j

swap 2 j

swap pivot 2 j

96 100 200 350 | 400 790 900 890 760
i i i j

96 100 200 350 400 | 790 760 890 900
i i j j

j j

swap pivot 2 j

96 100 200 350 400 760 790 890 900

i j a[i] a[j] a[pivot]

0 5 350 96 350

2 4 400 100 350

4 3 400 200 350

6 2 200 100 200

1 0 96 200 200

0 1 200 96 200

0 2 100 200 100

2 1 200 96 100

4 8 400 760 400

5 8 790 900 720

7 6 890 900 790

2 8 890 900 890